IGCSE 0607 Practice Questions — E4.3

Length and midpoint

Total Questions: 6 — **Total Marks:** 42 Marks

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1 Point A has coordinates $(1, 4)$ and point B has coordinates $(7, -4)$.

(a) Calculate the coordinates of the midpoint of the line segment AB.

(..... , ...) **[2]**

(b) Calculate the length of the line segment AB.

..... **[3]**

2 A circle has diameter AB, where A is the point $(-1, 2)$ and B is the point $(5, 8)$.

(a) Find the coordinates of the centre of the circle, C.

(..... , ...) **[2]**

(b) Find the exact radius of the circle.
Give your answer in the form $a\sqrt{b}$, where a and b are positive integers and b is as small as possible.

..... **[3]**

(c) Point D is such that the midpoint of CD is $M\left(\frac{5}{2}, \frac{3}{2}\right)$.

Find the exact length of AD, giving your answer in the form $c\sqrt{d}$, where c and d are integers and d is as small as possible.

..... **[5]**

3 A circle has its centre at the origin, $O(0, 0)$.
The point $P(-3, 4)$ lies on the circumference of this circle.

(a) Work out the radius of the circle.

..... **[2]**

(b) The line segment PQ is a diameter of the circle.
Find the coordinates of Q.

(..... , ...) **[2]**

(c) The point R has coordinates $(k, \sqrt{11})$.
R also lies on the circumference of the circle, and k is a negative number.
Work out the exact value of k.

$k =$ ..... **[3]**

4 Point P has coordinates $(-3, 5)$ and point Q has coordinates $(k, -1)$.
The length of the line segment PQ is 10 units.

(a) Find the two possible values of k.

$k =$ ..... , $k =$ ..... **[4]**

(b) Given that $k > 0$, point M is the midpoint of PQ.
Find the coordinates of M.

(..... , ...) **[2]**

5 The points A and B have coordinates $(-2, 5)$ and $(4, -3)$ respectively.

(a) Calculate the exact length of the line segment AB.

..... **[3]**

(b) Find the coordinates of the midpoint of AB.

(..... , ...) **[2]**

6 The points $A(2p, p)$ and $B(3p, 4p)$ are joined by a straight line, where $p > 0$.

(a) The length of the line segment AB is $5\sqrt{10}$.
Find the value of p.

$p =$ ..... **[4]**

(b) Point C lies on the line segment AB such that the ratio $AC : CB = 2 : 3$.
Using your value of p from part (a), find the coordinates of C.

(..... , ...) **[3]**

(c) The point $D(x, y)$ is such that $ABCD$ forms a rhombus.
Work out the coordinates of D.

(..... , ...) **[2]**